ED-Finder — the plan

Nov 8, 2026 · @Brian

Why we're rebuilding

We're rebuilding ED-Finder for its real scale: one invite-only group of 10–20 commanders, served from a Raspberry Pi at home.

The app started on a Pi. Pre-rating all 198.5 million systems grew it into a 1.1 TB database, a 192 GB server and weeks of capacity work. Hosting now costs £148 a month: £120 for the server, £16 for Contabo and £12 for backups.

The process grew to match. The repo carries 34 workflows, about 144k lines of tests and 94k lines of docs. Tests passed while the map looked broken, and merges stalled on reviewer quotas.

Principles from here:

- Build for 10–20 friends, not for 198 million systems.
- Keep the whole raw galaxy, and score areas on demand near the frontier.
- One box at home runs the app. Pay for nothing a home machine can do.
- Copy what works into a new repo. Don't rewrite it.
- Tests produce evidence you can watch: videos and screenshots.
- Keep the process sized to the group.

What we're building

One app that combines EDColonise, EDSM, Raven Colonial (planning and hauling) and BGS tools, for an invite-only group that all uses EDMC.

The journey runs Find → Look up → Plan → Haul, with BGS alongside and the map underneath.

Section	Replaces	What it does	Today
Find	EDColonise	Claimable systems near the live frontier, ranked by colony type, hiding claimed or colonised systems, sorted by distance from you or your carrier	Ranking built. Availability and the live frontier missing
Look up	EDSM	System, body and station pages: factions, economy and colonisation status. Your private travel log, and optionally where the group is	Inspect and journal import built
Plan	Raven (planning)	Layout, slots, economy, build order, Raven import and export	Built through S5c
Haul	Raven (hauling)	Construction needs against deliveries, who hauled what, carrier stock, shopping list, nearest seller, deadline countdown, team view	Not built
BGS	BGS-Tally and EliteBGS	Faction influence history and states, warnings, tick times, the group's activity per tick, a Discord report	Not built
Map	The in-game galaxy map	Galaxy overview, local stars, colonisation markers, click through to Look up and Plan	Built, but looks poor. Top priority after the move

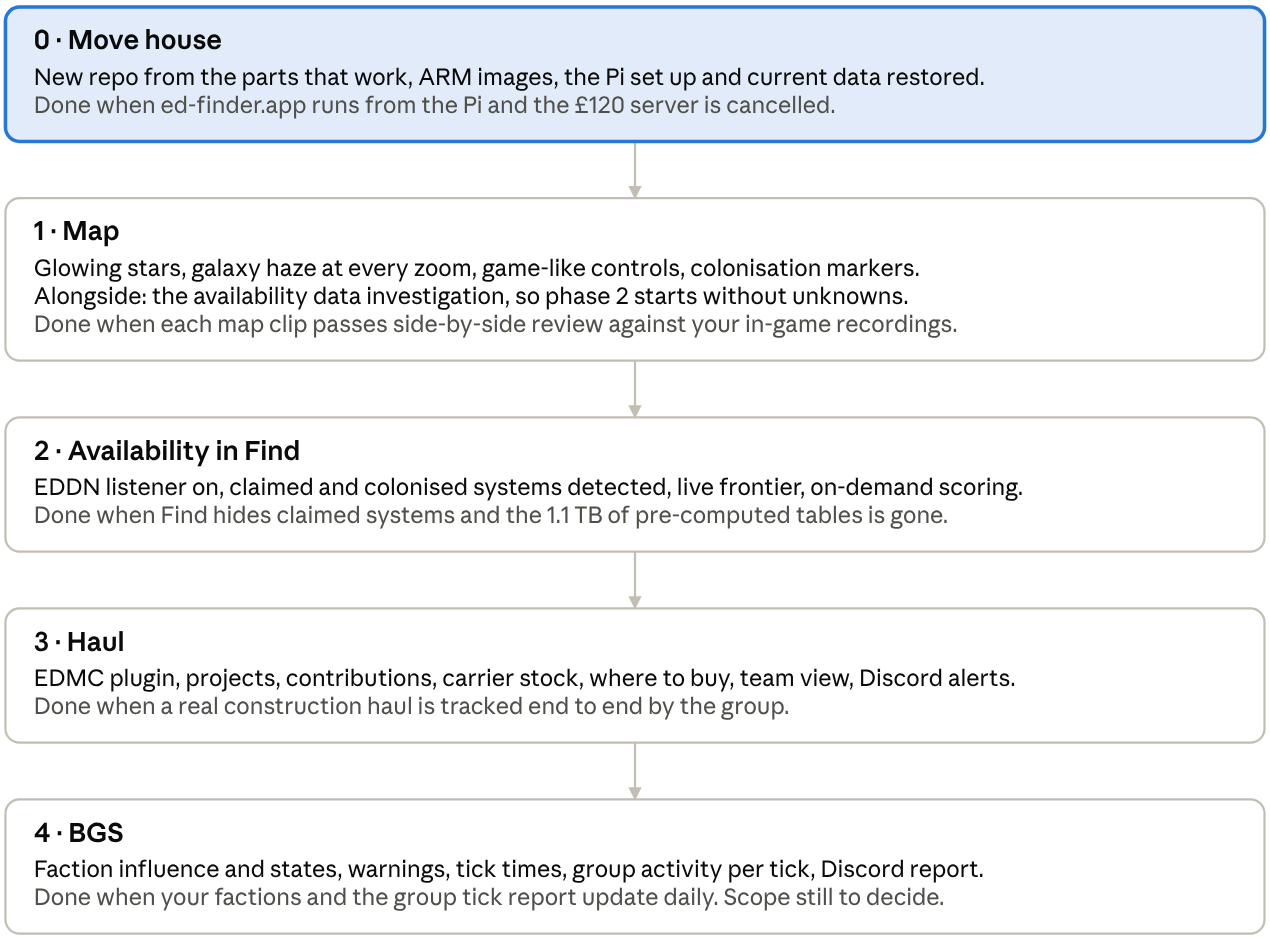
Access is invite-only: Frontier sign-in plus an invite list you manage. Projects, plans and BGS views are shared with the group by default. Travel logs and raw journal events stay private unless shared.

Claims start only from core systems and finished player colonies. The Mega Guide (v2.3) says Colonia's original systems can't be claimed from, so the frontier follows player bridges.

Order of work

The move comes first and adds no features. After it, the map leads, then availability in Find, Haul and BGS.

Move house first, then the map leads the features



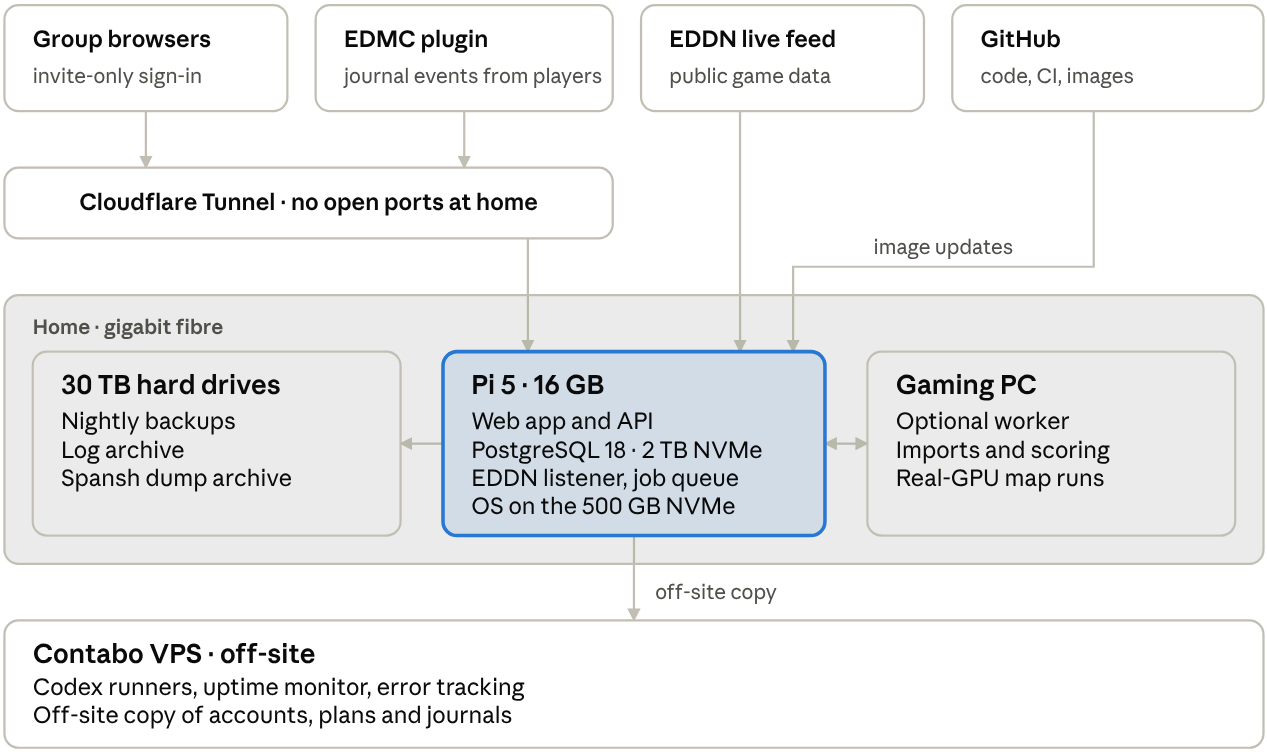
order of work · 5 phases, each with its done-when

Each phase is usable on its own, so the plan can pause or reorder after any of them.

Hosting

The app runs on one Raspberry Pi at home, Contabo keeps the tooling, and the £120 server goes after the move.

Everything runs at home; only tooling and a backup copy live off-site



hosting · Pi at home, Contabo off-site

Visitors and the EDMC plugin reach the Pi through Cloudflare Tunnel, so no ports open at home. The Pi pulls EDDN data and GitHub images itself.

Monthly cost	Today	After the move
Main server	£120	£0 (the Pi, plus electricity)
Contabo VPS	£16	£16
Backups	£12	£0–5 (Pi drives, plus a small off-site copy)
Total	£148	about £16–21

Pi drive	Holds
500 GB NVMe	Operating system, Docker, images, logs
2 TB NVMe (to buy)	PostgreSQL data: today's 1.1 TB, about 630 GB after on-demand scoring

Pi drive	Holds
30 TB USB hard drives	Backups, log archive, Spansh dumps

Both NVMe slots in the Pironman 5-MAX share one PCIe 2.0 lane. That's fine for serving pages; only the one-off 1.1 TB restore will be slow. The raw galaxy measured 616 GB on 30 October, plus 15 GB of density data.

Data sources

Seven sources cover everything, and only one input is unproven: whether a system can still be claimed.

Source	What it gives us	Used by	Status
Spansh galaxy dumps	Every system, body and station	Find, Look up, Plan	Imported
EDDN live feed	Population and factions (FSDJump and Location), stations (Docked), commodity markets	Availability, BGS, where to buy	Listener written, never switched on
Our EDMC plugin	Construction needs and deliveries, contributions, cargo, carrier transfers, missions, bounties, trade, travel log	Haul, BGS activity, Look up	New
Frontier sign-in and CAPI	Identity. Possibly carrier cargo, if EDMC can pass it to plugins	Accounts, Haul	Sign-in built
CRE rulebook	Game mechanics values with stable rule IDs and sources	Find scoring, Plan	v0.5.0 pinned
CMDR Zoy's tick detector	The daily tick time, as BGS-Tally uses	BGS	New
EDSM API	One-off backfill of faction influence history	BGS first load	Optional

EDMC already sends each player's data to EDDN, so the group's own flying keeps its systems fresh.

The unproven input: the 7 Nov research found the importer reads `isColonised` and `isBeingColonised`, but Spansh's dumps don't carry those fields. EDDN has no colonisation schema either. A read-only investigation runs during the map phase:

1. Count colonised against populated systems in the database.
2. Check what EDDN reveals about construction sites and newly populated systems.
3. Compare with Spansh, then choose the source for "still claimable".

Software stack

We keep the stack that works and drop what only served a remote server or whole-galaxy rebuilds.

Keep:

Layer	Choice	Why
Frontend	SvelteKit 2, Svelte 5, TypeScript	Works today; the whole live app is built on it
Map renderer	Babylon.js	Boxed into one folder (about 3,800 lines). Rendering runs in each visitor's browser
API	Python 3.14, FastAPI, asyncpg, generated TypeScript client	The scorer and importer are already Python
Database	PostgreSQL 18, history tables split by month	Spatial indexes, job queue and notifications in one place
Background work	PostgreSQL job queue (the existing chunk pattern), LISTEN/NOTIFY for live Haul updates	No extra services
Live feed	EDDN listener (`apps/eddn_v3`)	Already written
Companion	EDMC plugin	The whole group already runs EDMC
Containers	Docker Compose, images built for x86 and ARM	One compose file on the Pi
Error tracking	The API's existing Sentry SDK, pointed at Sentry's free tier or GlitchTip on Contabo	Already wired; needs only a DSN
Redis	Only if the API still needs it after the move	Otherwise PostgreSQL covers it

Drop:

Item	Why it goes
Whole-galaxy pre-computation	About 470 GB per generation and a 1.6 TB rebuild peak; on-demand scoring replaces it
NATS	Relays a single importer event; PostgreSQL already holds the truth
Blue/green slots	Each API slot may use 4 GB and each web slot 1 GB, doubled on a 16 GB Pi. A few seconds of downtime is fine
Sealed receipts and governance workflows	Built for agents operating a remote server over SSH. The Pi has no open ports, and many serve whole-galaxy rebuilds
Prometheus and Grafana	Not running in V3 production; uptime checks, logs and a health page cover it
GlitchTip server on the Pi	Five containers; error tracking lives elsewhere
React tree and V2 routers	Retired; not built or tested

Open-source code worth reusing, with each licence checked, is on the `Open-source shortlist` tab.

The new repo

A new repo is copied from the parts that work, and the old repo is archived read-only as reference.

What	Size	Verdict
`apps/web` (Svelte app and its tests)	~106k lines	Keep: this is the product
`apps/api`	~53k lines	Keep the V3 routers: search, archetypes, Ratings V4, plan, map, systems, journal, auth, watchlist. Leave V2 leftovers such as `colony_planner`, `optimiser`, `warehouse_planner_evidence`

What	Size	Verdict
`apps/importer`, `packages`, `sql/v3`	~77k lines	Keep the V3 pieces. The schema arrives as one starting file matching production
`tests`	~144k lines	Keep tests for the code that comes across
`frontend/` (retired React app)	~72k lines	Leave
`scripts/` (incl. ~37k of operator tooling)	~67k lines	Leave almost all
`docs/`	~94k lines	Leave. Start with a short README, this plan and a one-page CLAUDE.md
GitHub workflows	34	Keep 3–4: test, build images, deploy, Codex runner

Ground rules:

1. Copy, don't rewrite. Improvements come later, one at a time.
2. The database stays exactly as it is; only the code around it shrinks.
3. Archive the old repo read-only. Its history, PRs and research stay available.
4. Re-create secrets by hand: GitHub secrets, the Frontier app credentials, Cloudflare. No secret files are copied.
5. Merge the green UI PRs into the old repo first, so their work comes across.
6. Build the new repo as part of the Pi move, so nothing is built twice.

The new CLAUDE.md states the scale and the rules in one page: 10–20 users, one Pi, copy don't rewrite, and which test tool owns what.

Test environment

Every PR produces a review pack of videos and screenshots, so you review a change by watching it, not by clicking through it.

What the old setup missed, from the 1 Nov review:

- All 1,129 frontend tests passed while every star rendered blue; colour tests used idealised star types.

- The map check shrank a screenshot to 128×64 and passed on more than ten non-background pixels.
- CI ran Chrome with the GPU disabled, so no test saw the map on real graphics hardware.
- Two browser lanes used different data, and neither matched production.
- No backup restore had ever been rehearsed.
- Unrelated flaky tests failed PRs and trained everyone to ignore red.

Tools, one job each:

Tool	Job	Runs
Cypress	Pass/fail user journeys and functional map checks, with video and screenshots	Every PR, in CI
Playwright	Map recordings with real mouse and keyboard input on a real GPU, traces, frame timing, side-by-side clips	Your gaming PC, when you start it
Vitest	Unit and component tests; map components in real-browser mode	Every push

Rule for every session: Cypress = pass/fail journeys in CI. Playwright = map recordings on the real GPU, plus the Vitest browser provider. A test lives in exactly one of them.

The pieces:

- **Golden data slice:** a real, versioned extract of the database. It covers the Bubble, a colonised frontier, Colonia, a dense core sample, nebulae and black holes, with private data removed.
- **One test stack:** the production compose file plus test data, started with one command, locally or in CI.
- **Replays:** a recorded day of real EDDN messages, and real journal sessions from the group, played back at speed through the listener and the plugin.
- **Review pack:** one page per PR with journey videos, map clips at 1760×988 beside your in-game recordings, screenshots with diffs, and traces for failures.
- **Staging on the Pi:** the real ARM build at a staging address, checked before every release.

- **Restore test:** weekly, last night's backup is restored into a scratch database and checked.

When	Where	What
Every push	GitHub runners	Unit, component, type checks, API and plugin tests
Every PR	GitHub runners	Full stack with the golden slice, all journeys on video, screenshot comparisons, map checks, replays, review pack
On demand	Gaming PC	Map clips on the real GPU, frame timing, side-by-side with the game
Before release	Pi staging	Quick journeys against the ARM build
Weekly	Pi	Restore test and backup and listener health checks

Rules learned the hard way:

1. Fixtures come from real data only: captured API responses and database extracts.
2. Map checks test meaning: star classes differ, markers show, no hard edge on the star cloud, no empty frames while zooming.
3. The app is built to be tested: a renderer-ready signal instead of fixed waits, a test-mode camera, seeded randomness and a fake clock.
4. Flaky tests are quarantined and fixed, never retried until green.
5. One harness, tiered by speed, with one set of data.

Running it

You run updates yourself with one script; backups are tested every week; and you hear about outages before your friends do.

Job	How
Build	GitHub Actions builds x86 and ARM images and tags each release
Deploy	An update script on the Pi pulls a tagged version, backs up first if it carries a migration, runs `docker compose up -d` and writes a deploy log

Job	How
Rollback	Run the script with the previous tag
Migrations	Reviewed SQL only, always after a fresh backup
Backups	Nightly database dump and log archive to the 30 TB drives. A small encrypted copy of the irreplaceable data goes off-site
Restore test	Weekly and automatic, into a scratch database
Uptime	Uptime Kuma on the Contabo box alerts you if the site goes down
Errors	Sentry's free tier, or GlitchTip on Contabo, via the API's existing SDK
Logs and health	Structured logs on the Pi, plus a health page for disk, memory, temperature and database size

Irreplaceable data means accounts, plans, journal imports, Haul history and deploy settings. The raw galaxy can be re-imported from Spansh, so its copy stays on the hard drives.

Pi setup notes:

- Leave the Pironman 5-MAX's RAID option off: the two NVMe drives do separate jobs.
- Use the recommended 27 W power supply, and give the hard drives their own power.
- Tune PostgreSQL for 16 GB (roughly 4 GB of its own cache). Today's settings assume 192 GB.
- Restore with PostgreSQL's dump and restore tools. Don't copy raw data files from the x86 server.

Next steps and open decisions

The only deadline is the server renewal, due about a week after 8 November; everything else can start when you're ready.

- [ ] Pay the current server for one more month
- [ ] Merge the green UI PRs: #1016, #1017, #1019–#1023, #1025 and #1032
- [ ] Re-run the flaky jobs on #1014, #1015, #1018 and #1036

- [ ] Restore Codex review, or merge with your OK while its quota is out
- [ ] Close the stale PRs #1034, #780, #718, #715 and #778
- [ ] Buy a 2 TB NVMe for the Pi and set it to boot from the 500 GB drive
- [ ] Create the new repo on GitHub
- [ ] Spike the Almanac Galaxy-Map in the new repo with our colonies and candidates (one day)
- [ ] Point a copy of BGS-Tally at a stub server and confirm colonisation events arrive

Open decision	Options
BGS scope	Your factions and watched systems, or any faction anywhere. Yours is much lighter on the Pi
Off-site full-database copy	Pi hard drives only, or add a small 1 TB storage plan
Spansh dump	Confirm it can still be downloaded, so the raw galaxy can be rebuilt
Group journals	Will the group share play sessions, with private details stripped, as replay test data?
Browsers	Which browsers the group uses. Chrome or Edge only shrinks the test matrix
Real-GPU runs	Happy to run the Playwright map suite on your gaming PC when you choose?
Galaxy map	The Almanac Galaxy-Map (looks like the game, non-commercial licence) or keep building on Babylon. Decide after the spike. If we choose it, Babylon goes too: the System Map is flat, so SVG can draw it
Companion plugin	Start on BGS-Tally's API, so nothing new goes on the group's PCs, or write our own EDMC plugin first

Decision log

Newest first. Add a row whenever a decision changes the plan.

Date	Decision	Detail
2026-11-08	Codex stays a required review	210 findings on 57 of 225 PRs since 1 October, about 93% fixed. If Codex is unavailable, your OK is enough to merge
2026-11-08	Octopus dropped	17 findings across 141 reviewed PRs since 1 October, mostly wrong or minor. Paid per review plus a hosted stack
2026-11-08	Test environment redesigned	Review pack with videos, golden data slice, replays, Pi staging, weekly restore test
2026-11-08	Cypress stays; Playwright only records	The earlier move away from Playwright stands. Playwright is limited to real-GPU map recordings and Vitest's browser mode
2026-11-08	Stack trimmed	Drop NATS, blue/green, receipts and governance workflows. No Prometheus or Grafana. No GlitchTip server on the Pi
2026-11-08	Invite-only, EDMC companion	Frontier sign-in plus an invite list. Data arrives through our EDMC plugin
2026-11-08	Priority after the move	Map, then availability in Find, then Haul, then BGS
2026-11-08	Product scope	EDColonise, EDSM, Raven planning and hauling, and BGS in one app
2026-11-08	New repo	Copy what works, don't rewrite, archive the old repo
2026-11-08	Pi 5 hosts the app	Pironman 5-MAX, 500 GB boot and 2 TB data NVMe, 30 TB of hard drives for backups, Cloudflare Tunnel
2026-11-08	Contabo keeps the tooling	Codex runners, off-site copy, uptime monitor
2026-11-08	Keep the raw galaxy, score on demand	Whole-galaxy pre-computation goes
2026-11-08	Leave the £120 server	Pay one more month, then cancel after the move